Session plan 6: Causal inference

Thu 15.10.26, 14:15–15:45 · Room 3.B48 · Basic reading: BF ch. 2–3 (pp. 13–52); additional: KKV ch. 3 · Applied reading: Alrababa'h, Marble, Mousa & Siegel (2021), *Can exposure to celebrities reduce prejudice?* · Presenters: **Stephanie Lindegger, Leonie Schär**

Learning goals: students can (1) name three uses of a correlation (description, forecasting, causal inference) with the age-turnout example; (2) explain why r can miss a relationship; (3) name four patterns why a correlation need not be causal (confounding, reverse causation, selection, chance); (4) define a causal effect with potential outcomes; (5) explain the fundamental problem of causal inference and why every design substitutes for a missing counterfactual.

Before class: presentation slides in OLAT (due Wed 14.10, 12:00)? · seminar questions 6 uploaded? · grading sheet `session06_handwritten.pdf` ready · board/whiteboard pen for the attendance list · deck in speaker view (press S)

Checkpoints: 14:30 start "Why correlation is not causation" · 14:44 start "Causation" · **15:07 end group exercise** · **15:08 presentation starts** (25 min max., then discussion until 15:42)

Time	Slide	Content	Interaction	Guiding questions	If behind
14:15–14:19	2	**Hook**: attendance and grades (hypothetical)	Show of hands; collect "no" reasons on the board	Should attendance be mandatory? Why not?	
14:19–14:25	3	Correlation = features that tend to occur together (BF p. 13); **older people vote more** (BF Fig. 2.4): describe, predict, explain (BF pp. 19–24)	Ask first, then reveal	What can we do with this fact? Does getting older *make* people vote?	
14:25–14:28	4	What r does not tell you: six simulated panels; linearity (BF pp. 30–31)	Question	Is r 0 evidence of no relationship?	1 min
14:28–14:30	5	Ice cream and drownings	Students explain	Does ice cream make people drown?	
14:30–14:34	6	**Four patterns**: confounding, reverse causation, selection, chance; calculus and college (BF pp. 23–24)	Map the board list	Which board reason is which?	
14:34–14:44	7	**Pairs**: police and crime, chocolate and Nobel prizes, old buildings; bonus: admission, grades and test scores (hypothetical)	6 min pairs, 4 min debrief	Which pattern fits each?	Debrief only C (selection: survivors)
14:44–14:47	8	Causal effect (BF pp. 37–38); "does X change Y?", not "the cause" (BF pp. 42–43)	Input		
14:47–14:50	9	Potential outcomes Y(1), Y(0), (BF p. 39)	Question	Which of your two outcomes will you see?	
14:50–14:53	10	**Fundamental problem** (BF p. 41); "We can't"	Students propose	What would you need to observe to know whether studying helped *you*?	
14:53–14:59	11	**Which comparison would you trust?** Before/after, self-selected, coin flip (BF pp. 41–42)	Vote, then problems with each	What could go wrong with A, B, C?	Vote only
14:59–15:01	12	Back to attendance; sessions 8–10	Plenary	What would you need to claim attendance *causes* grades?	Skip

Time	Slide	Content	Interaction	Guiding questions	If behind
15:01–15:07	13	**Group exercise** on seminar questions 6	Pairs, 1–2 share	Is it the right counterfactual? How to approximate it?	No sharing; **hard stop 15:07**
15:07–15:08	14	Hand over			
15:08–15:33	—	**Student presentation** (25 min max.): summary and critique ~10 min, alternative design ~15 min	Presentation; grade with the sheet		Signal at 20 min
15:33–15:42	—	**Discussion** to improve the proposal (prompts below)	Plenary	See prompts below	
15:42–15:45	15–18	Session 7 at 16:15, same room (Kleinberg et al. 2015; Hofman et al. 2021; Flückiger & Kohm); take-home exercise I goes online in session 7; R track	Admin		

Discussion prompts (if the room is quiet): Salah is Muslim *and* very successful: which of the two does the paper's design isolate? · Synthetic control for Merseyside: what else changed in Liverpool around 2017 that could also affect hate crimes? · Tweets of Liverpool fans vs. fans of other clubs: are they comparable, and who tweets at all (selection)? · Which counterfactual does each of the three designs (hate crimes, tweets, survey experiment) construct? · Does the effect travel to celebrities who are less successful or less liked?

Grading (0–5 each, 15/30 = pass): clarity · understanding · critique quality · reasoning · feasibility · discussion. Both presenters, one grade.

Notes during class: